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Public Review for

Efficient Identification of Uncongested Internet Links for Topology Downscaling

Fragkiskos Papadopoulos & Konstantinos Psounis

I copy the paper summary from one of the reviews: "The paper presents an efficient scheme to identify uncongested links in a given network. The primary motivation is to complement studies on topology downscaling: once uncongested links are identified, they can be safely ignored. In addition, the identification of uncongested links is also useful for other purposes such as traffic engineering and capacity planning. The proposed scheme is based on a known result about the queuing delay distribution from large deviation theory. The authors demonstrate how to utilize that result when only flow-level (instead of packet-level) statistics are available (say from NetFlow). In the process, they also derive a simpler formula to obtain the variance of a packet-level arrival process from flow-level statistics. The proposed scheme is validated using ns-2 simulations."

All three reviewers liked the paper, agreeing that it focuses on an important problem and that it offers a nice and novel solution. On the other hand, they also agree that the paper lacks on validation. The proposed mathematical model makes a number of assumptions, some of them difficult to accept. Unfortunately, even though the authors could have validated at least some aspects of the proposed methodology experimentally, they only relied on simulations.

For instance, the paper claims that the Hurst parameter of the traffic on a link can be estimated online from the observed flow size distribution at that link (Section 6.2). This is easier said than done. The accurate estimation of the flow size distribution tail parameter requires a significantly large sample, which then raises concerns about whether the traffic remains stationary in that interval. The authors could have examined this issue experimentally, using publicly available packet traces.

The reader should be also alarmed about the assumptions behind the main mathematical result of the paper (Theorem 1). The result is proved for homogeneous TCP flows (same RTT, same maximum window size). Further, note that even though that particular link may be uncongested, it may carry flows that experience packet losses elsewhere. In that case, the TCP throughput derivations in the proof of Theorem-1 may not hold, at least for those flows.

At the end of the day, the reader is the ultimate "consumer" and the reviewer that matters the most. Enjoy reading the paper and do not hesitate to post your comments at the online version of CCR.

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Efficient Identification of Uncongested Internet Links for Topology Downscaling

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ABSTRACT

It has been recently suggested that uncongested links could be completely ignored when evaluating Internet's performance. In particular, based on the observation that only the congested links along the path of each flow introduce sizable queueing delays and dependencies among flows, it has been shown that one can infer the performance of the larger Internet by creating and observing a suitably scaled-down replica, consisting of the congested links only. Given that the majority of Internet links are uncongested, it has been demonstrated that this approach can be used to greatly simplify and expedite performance prediction.

However, an important open problem, directly related to the practicability of such an approach, is whether there exist efficient and scalable ways for identifying uncongested links, in large and complex Internet-like networks. Of course, such a question is not only very important for scaling down Internet's topology, but also in many other contexts, *e.g.* such as in traffic engineering and capacity planning.

In this paper we present simple rules that can be used to efficiently identify uncongested Internet links. In particular, we first identify scenarios under which one can easily deduce whether a link is uncongested by inspecting the network topology. Then, we identify scenarios in which this is not possible, and propose an efficient methodology, based on the large deviations theory and flow-level statistics, to approximate the queue length distribution, and in turn, to deduce the congestion level of a link. We also demonstrate how simple commonly used metrics, such as the link utilization, can be quite misleading in classifying an Internet link.

Categories and Subject Descriptors

C.2.5 [**Local and Wide-Area Networks**]: Internet; C.4 [**Performance of Systems**]: Measurement techniques, Modeling techniques; G.3 [**Probability and Statistics**]: Queueing theory, stochastic processes; C.2.1 [**Network Architecture and Design**]: Network topology

General Terms

Performance, Measurement, Theory

Keywords

Topology downscaling, uncongested link identification

1. INTRODUCTION

Understanding the behavior of the Internet and predicting its performance are important research problems. These problems are made difficult because of the Internet's large size, heterogeneity and high speed of operation.

Researchers use various techniques to deal with these problems: modeling, *e.g.* [20, 3, 13, 28], measurement-based performance characterizations, *e.g.* [11, 33, 19, 35, 22], and simulation studies, *e.g.* [1, 25, 41, 18]. However, these techniques have their limitations.

First, the heterogeneity and complexity of the Internet makes it very difficult and time consuming to devise realistic traffic and network models. Second, due to the increasingly large bandwidths in the Internet core, it is very hard to obtain accurate and representative measurements. And further, even when such data are available it is very expensive and inefficient to run realistic simulations at meaningful scales.

To sidestep some of these problems, Psounis et al. [29] have introduced a method called SHRiNK, that predicts network performance by creating and observing a *slower* downscaled version of the original network.[1] In particular, SHRiNK downscales link capacities such that, when a sample of the original set of TCP flows is run on the downscaled network, a variety of performance metrics, *e.g.* the end-to-end flow delay distributions, are preserved.

This technique has two main benefits. First, by relying only on a sample of the original set of flows, it reduces the amount of data we need to work with. Second, by using actual traffic, it short-cuts the traffic characterization and model-building process. These in turn, expedite simulations and experiments with testbeds, while ensuring the relevance of the results. However, this technique did not solve the very important problem of having to deal with large and complex network topologies, like the Internet topology.

With the above problem in mind, Papadopoulos et al. [31, 32] proposed two methods that can be used to scale down the topology of the Internet, while preserving the same *performance* metrics and having the same benefits with SHRiNK. [2] In particular, by defining a link to be congested if the link imposes packet drops or significant queueing delays, it has been shown that it is possible to infer the performance of the larger Internet by creating and observing a suitably scaled-down replica, consisting of the congested

[1] SHRiNK: Small-scale Hi-fidelity Reproduction of Network Kinetics.

[2] The methods are called DSCALEd (Downscale using delays), and DSCALEs (Downscale using sampling).

links only. Further, based on the observation that the majority of backbone links are uncongested [3, 4, 12, 10] it has been demonstrated that these techniques can be used in practice, to dramatically simplify and expedite performance prediction.

A main requirement of this approach is that uncongested links are known in advance. However, while links that cause packet drops can be easily detected by a monitoring tool (*e.g.* via standard SNMP techniques, or more sophisticated network tomography techniques), measuring the queueing delays on every other link to determine whether these are negligible, is clearly not a scalable procedure. Further, it becomes critical in high-speed backbone routers [3, 33, 34]. Hence, to make topological downscaling more practical, we need efficient ways to identify uncongested links, without having to explicitly measure their delays. This is the main contribution of this paper.

In particular, in this paper we present an efficient and scalable procedure to identify which of the links of a network topology that do not impose packet drops are uncongested, i.e. they do not impose significant delays either. Our procedure consists of rules under which one can easily identify uncongested links by inspecting the network topology, and whenever this is not possible, by efficiently using a known model from the large deviations theory (based on Fractional Brownian Motion), to approximate the queue length distribution. We further demonstrate that a commonly used metric, the link utilization, can be quite misleading in assessing of whether a link is congested; Internet links at high utilizations, *e.g.* 90% or even higher, may still be uncongested, if both the degree of statistical multiplexing and their capacity are large.

The large-deviations model we use requires knowledge of *packet-level* statistics at the link of interest. In particular, it requires knowledge of the average packet arrival rate λ, of the variance of the arrival process σ^2, as well as of the Hurst parameter H, an index of long-range dependence in the arrival process [14]. However, as with the queueing delays, it is difficult and not scalable to estimate these parameters by monitoring packets on every link of interest. In our approach, we make efficient use of this model in the sense that we choose to *infer* these parameters from *flow-level* information at the link of interest. We have chosen to do this based on the observation that it is much easier to monitor flows on a router, instead of packets [3, 4]. This argument is further strengthened by the fact that information on flows can be either collected on the link we want to study or at the edges of a backbone network. Combining flow and routing information at the edge routers will give us information on each link of the network that we want to study [3, 4]. This alleviates the burden of having to monitor many links and makes the measuring procedure scalable. [3]

However, while simple expressions connecting λ and H to flow-level statistics exist, *e.g.* [3, 39], inferring σ^2 from flow-level information is much more involved. Another contribution of this paper is that we derive a new expression for σ^2. What distinguishes our expression from earlier ones [3, 4, 16] is that it requires less flow-level information, and it has been derived without any assumptions, by explicitly taking into consideration the TCP feedback mechanism and long-range dependence.

[3]Tools, *e.g.* such as NetFlow, can be easily utilized to provide flow-level information on backbone routers [24].

While our main motivation in this paper is to complement the work on topology downscaling, by efficiently identifying the uncongested links that can be ignored, our approach is quite general and can be used beyond this context, *e.g.* by network operators and managers for traffic engineering and capacity planning.

The rest of the paper is organized as follows. In Section 2 we briefly review the main concept of performance-preserving topological downscaling. In the Section 3 we identify the scenarios under which one can easily deduce whether a link imposes negligible queueing, by inspecting the topology. Whenever this is not possible, we review in Section 4 the large-deviations model that we will be using to approximate the queue length distribution. In Section 5 we explicitly identify the conditions that should hold in the context of TCP networks for this model to be valid. In Section 6 we infer the packet-level information required to use the model, from flow-level information. In Section 7 we validate the model and our theoretical arguments using simulations with TCP traffic. In the same section we also present experiments using the CENIC backbone [5], to demonstrate how the model can be used in practice to identify uncongested links, and to decide which links to ignore when performing topological downscaling. Comparison with earlier work follows in Section 8, and we conclude in Section 9.

2. PERFORMANCE-PRESERVING TOPOLOGICAL DOWNSCALING

In this section we briefly review the main concept of downscaling TCP networks. For more details, the interested reader is referred to [32, 31].

Before proceeding, let's first review the definition of an "uncongested" link in the context of downscaling. An uncongested link is a link which: (i) does not impose any packet drops, and (ii) its queueing delays are negligible compared to the total end-to-end delays of the packets that traverse it, *e.g.* one order of magnitude smaller. The *majority* of backbone links have both of these properties. In particular, it is well documented that the end-to-end delay inside a backbone network is dominated by the propagation delay, and that most of the backbone links never impose packet drops [12, 10, 34, 3, 4, 33, 11]. [4] The main idea in downscaling is to reduce the topology of the network by ignoring uncongested links.

As an illustrative example, let's consider the topology shown in Figure 1. In this topology we can see two congested links, and two groups of flows, Grp1 and Grp2. [5] Observe that Grp1 traverses only one congested link, whereas Grp2 traverses both.

In [31, 32] two methods have been proposed (DSCALEd and DSCALEs) that build scaled replicas consisting of the congested links only, along with the groups of flows that traverse them, which are called groups of interest. [6] For the example shown in Figure 1, the resulting scaled replica is shown in Figure 2. Then, the methods adjust the round-trip times in the scaled replica appropriately, such that the

[4]Congested links usually exist at access points, public exchange points, etc.

[5]A group of flows consists of those flows that follow the same network path.

[6]The scaled replica may also include uncongested links that we wish to study, and the groups of flows that traverse them.

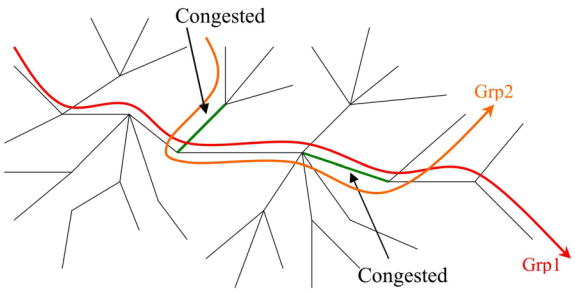


Figure 1: Original network.

performance of the replica can be extrapolated to that of the original network. (The benefits of using this approach over other performance measurement techniques and tools have been discussed in [32]).

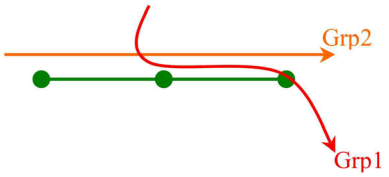


Figure 2: Scaled replica.

A main requirement of topology downscaling is that we know in advance which links of the original network are uncongested. However, as mentioned earlier, while links that cause packet drops can be easily detected by a monitoring tool, measuring queueing delays on every other link to determine whether these are negligible, is clearly not a scalable procedure, and becomes quite difficult in high-speed backbone routers [33, 34, 3]. Hence, for downscaling to be practical, we need efficient ways to identify links with negligible queueing, without having to explicitly measure their delays. (Notice that in the context of downscaling the queueing delay threshold under which a link is considered to be uncongested is relative to the end-to-end delays of the packets that traverse it. In other contexts, *e.g.* in traffic engineering, this queueing delay threshold may vary and it is an option of the network operator/manager. And, as we have mentioned earlier, the procedure that we will describe in this paper for classifying a link as uncongested is quite general and applicable beyond the context of downscaling.)

3. IDENTIFICATION OF UNCONGESTED LINKS BY TOPOLOGY INSPECTION

In this Section we identify the conditions under which one can decide whether a link is uncongested by just inspecting the network topology.

Our starting point is based on the observation that each link that belongs to the path of a group of flows of interest (*e.g.* the path of Grp1 in Figure 1), can be considered as being part of sub-topologies similar to those shown in Figures 3(i) ... 3(iii).

Now, let's study the conditions under which link Q_2 in Figures 3(i) and 3(ii) and link Q_1 in Figure 3(iii) impose insignificant queueing. (The C's in the plots correspond to capacities.) Let's first concentrate on the topology shown in Figure 3(i). Clearly if $C_1 \leq C_2$ there is not going to be any queueing at Q_2, whereas if $C_1 > C_2$ significant queueing at Q_2 is possible. Let's move to the topology shown in Figure 3(ii). If $\sum_{j=1}^{N} C^{1j} \leq C_2$ there is not going to be any queueing at Q_2, but if $\sum_{j=1}^{N} C^{1j} > C_2$ significant queueing at Q_2 is possible. Finally, for the topology shown in Figure 3(iii), if $C_1 \leq \sum_{j=1}^{N} C^{2j}$ we can have significant queueing at Q_1. But, if $C_1 > \sum_{j=1}^{N} C^{2j}$, the C^{2j}'s will regulate the arrivals at Q_1 (through the TCP feedback mechanism) and queueing, which is caused only by the first few packets of new unregulated flow arrivals, will be negligible.

Therefore, in summary, *the only case where one can decide by inspecting the network topology, that a link imposes negligible queueing, is the case where the link carries traffic from/to links for which the sum of their capacities is smaller than the capacity of the link.*

For the rest of the cases, we will make efficient use of a model from the theory of large deviations to approximate the queue distribution. We briefly review this model in the next section.

4. USING LARGE-DEVIATIONS THEORY TO APPROXIMATE THE QUEUE DISTRIBUTION

Consider a link/queue, and let $A(t) = A(0,t)$ denote the total traffic that has arrived at the queue (*e.g.* in units of packets or bits) in the interval $(0,t]$, with $t \in \mathbb{R}^+$. Further, let λ denote the average input (arrival) rate, and C denote the queue's service rate (link capacity). To ensure stability, we assume that $\lambda < C$.

We are interested in the steady-state probability $P(Q > \delta B)$ of the buffer content Q exceeding some prespecified level $\delta B > 0$, where $0 < \delta \leq 1$. Assuming an infinite buffer size, this probability can be expressed in terms of the arrival process $A(t)$, as follows (*e.g.* see [14]): $P(Q > \delta B) = P\left(\sup_{t \geq 0}[A(t) - Ct] > \delta B\right)$. This relation is oftenly used to approximate the corresponding probability in a system with finite buffer equal to B, when B is large [14].

Now, let's assume that the input process can be well described by a Fractional Brownian Motion process. That is, let's assume that $A(t)$ is a Gaussian process with mean $E[A(t)] = \lambda t$ and variance $\mathrm{Var}[A(t)] = \sigma^2 t^{2H}$, where $H \in [0.5, 1)$. (The constant H is the Hurst parameter. For $H = 0.5$ the process has independent increments, whereas for $H > 0.5$ the increments of the process are long-range dependent.) Finally, let $I(H) = \frac{(C-\lambda)^{2H}(\delta B)^{2-2H}}{2\sigma^2 K^2(H)}$, where $K(H) = H^H(1-H)^{1-H}$. Then, using large-deviations theory, it can be shown that the following relationship holds for $P(Q > \delta B)$ [14, 37]:

$$P(Q > \delta B) \leq \exp\left(-I(H)\right). \quad (1)$$

The above relation is known in the literature as the *large-buffer asymptotic* upper bound and the function $I(H)$ is called the *large-deviations rate function*. If B is sufficiently large, Equation (1) is used as an approximation of the queue length distribution. For $\delta \geq \frac{1}{B}$ a better bound/approximation

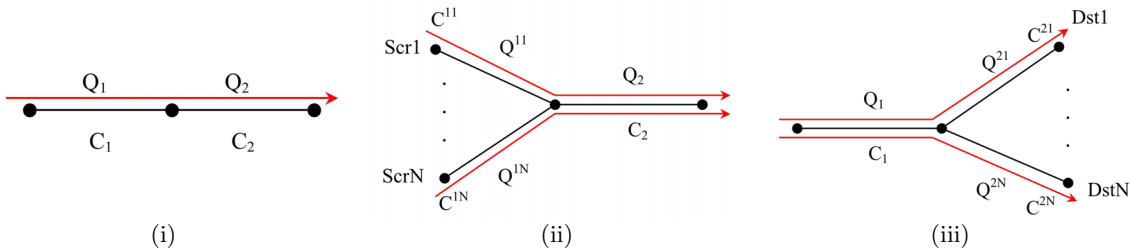


Figure 3: Toy network topologies used to illustrate when a link can be considered as uncongested by topology inspection.

is [23]:

$$P(Q > \delta B) \leq \frac{1}{(\delta B)^{\gamma}} \exp\left(-I(H)\right), \qquad (2)$$

where $\gamma = \frac{(1-H)(2H-1)}{H}$. Hence, for better approximating the queue distribution for any δ, one can take the minimum of Equations (1) and (2).

The effectiveness of this model has been demonstrated in the context of open-loop networks, *e.g.* [8, 26, 14], and has been used several times in the context of TCP networks, *e.g.* [6, 40, 7, 42, 34]. Next, we clearly identify the conditions under which the model is valid in this latter context. Then, we show how to use it efficiently, by inferring its parameters from flow-level information.

5. APPLICATION TO TCP NETWORKS

As we can see from the previous section there are two requirements for the model described there to be accurate: (i) the buffer size B should be large enough, and (ii) the input process should be well-described by a Gaussian process. We now briefly explain why both of these conditions hold true in the context of TCP backbone networks.

First, Internet routers today are still sized according to the rule-of-thumb, where the buffer size equals the bandwidth-delay product [15]. Since capacities in backbone links are quite large, so that they can support a large number of flows, the buffer size B is also large.

Further, while it is well known that if multiple TCP flows share a bottleneck link can get synchronized with each other [44, 9, 43], flows are not synchronized in a backbone router that carries a large number of them, with various round-trip, processing and startup times, even if significant amounts of packet drops occur. These variations are sufficient to prevent synchronization, and this has been demonstrated in real networks [2, 12, 17].

Under the assumption of a large number of desynchronized TCP flows, the evolution of the flow window sizes becomes loosely correlated, and the distribution of their sum can be well approximated by a Gaussian distribution. This is justified by the Central Limit Theorem (CLT), it is supported by empirical measurements, and it has been argued in several recent studies for sizing backbone routers [2, 7, 16].

Hence, requirements (i) and (ii) both hold true in the case of Internet backbone networks. Finally, notice that the model of the previous section also accounts for long-range dependence in the increments of the aggregate Gaussian input traffic, which is another well-known characteristic of traffic in the Internet, *e.g.* [42, 35, 39].

6. PARAMETER INFERENCE

Using the model of Section 4 requires knowledge of the *packet-level* statistics λ, σ^2, and of the parameter H. As mentioned earlier, it is difficult and not scalable to estimate these parameters by monitoring packets on every link that we want to study. As we have said, we prefer to monitor flows, which is much easier [3, 4]. Therefore, in this section we show how to infer these parameters from flow-level information. Before proceeding, recall that it is easy to detect links that impose packet drops, and thus we are interested in detecting which of the other links impose significant queueing delays.

To be able to infer the *packet-level* statistics λ, σ^2 and the parameter H at the link we want to study, it is necessary (and sufficient) to have the following *flow-level* information: (i) the flow size distribution $F(s)$ of the flows traversing the link, (ii) the average flow arrival rate at the link, which we denote by r, and (iii) the average and the variance of the number of active flows on the link, which we denote by $E[A]$ and $\mathrm{Var}(A)$ respectively. (As usual, we say that a flow is "active" on a link, if the link belongs to the path of the flow, and the flow has more data packets to send.) In practice, this flow-level information can be easily extracted from a router, *e.g.* using NetFlow [24]. [7]

6.1 Estimating λ

Let S be the random variable representing the size of a flow. Since we know $F(s)$ we can easily compute the average flow size $E[S]$. For links with no drops, an intuitive and well-known expression for λ (*e.g.* see [3]) is: [8]

$$\lambda = rE[S]. \qquad (3)$$

The relation above states that the average packet arrival rate is equal to the average arrival rate of flows times the average amount of load brought by each flow. Note, that for a system to be stable (in the sense that the number of active flows never grows to infinity) it is required that $\lambda = rE[S] < C$ [13]. We assume this to be the case here. (Recall that this condition is required in order to be able to

[7]Note that we are assuming time-intervals where traffic remains stationary, and therefore, where the flow-level statistics that we need do not change. It has been demonstrated in real networks that traffic remains (approximately) stationary whithin 30-minute intervals, *e.g.* [3, 4].

[8]This relation also ignores any TCP timeouts that may be caused due to unusual sudden increases in queueing delays. However, such timeout events are rare since TCP is usually quite efficient in adapting its retransmission timeout interval.

invoke the model of Section 4.) Next, we use another known result to show how one can estimate the Hurst parameter H.

6.2 Estimating H

The long-range dependence of Internet traffic has been shown to be the result of a heavy-tailed flow size distribution [39, 12]. A heavy-tailed distribution is one in which $P(S > s) \sim s^{-\alpha}, 1 < \alpha < 2, \text{as } s \to \infty$.

At large time-scales, *e.g.* greater than the average round-trip time, the Hurst parameter H is directly related to the parameter α (called the shape parameter) of the size distribution. According to [39]:

$$H = \frac{3-\alpha}{2}. \tag{4}$$

Since we know the flow-size distribution $F(s)$ (and hence its shape parameter α), we can use Equation (4) to approximate H. (Also, note that there exist many methods for accurately estimating the shape parameter of a heavy-tailed distribution from a finite and not very large number of samples *e.g.* see [39].)

6.3 Estimating σ^2

To date, only few studies exist that relate σ^2 to flow-level information [3, 4, 16]. However, these studies either make unnecessary simplifying assumptions on how flows transmit their packets [16], or give fairly complicated expressions that require more information and measurements [3, 4], than we actually need.

Since we are interested in links with no drops, it turns out that we can derive a new simpler expression for σ^2, assuming knowledge of only the flow-level information mentioned earlier, and without making any assumptions on how flows transmit their packets. (For a detailed comparison with prior work see Section 8.) The expression is given in the following Theorem: [9]

THEOREM 1.

$$\sigma^2 = \frac{E[A]\,Var(W) + (E[W])^2\,Var(A)}{(E[RTT])^{2H}}, \tag{5}$$

where $E[W]$ is the average congestion window size of a flow that traverses the link and $Var(W)$ its variance, $E[RTT]$ is the average round-trip time of a flow, and $E[A]$, $Var(A)$ are respectively the average and variance of the number of active flows on the link.

PROOF. Assume that time is slotted with the duration of slot i be equal to the current round-trip time. Further, let the current round-trip time be the same for all flows traversing the link. Now, denote by P the total number of packets that arrive to the link/queue within some time-slot. Then, $P = \sum_{j=1}^{A} W_j$, where A is the random variable representing the number of active flows in a time-slot, and W_j is the random variable representing the congestion window size of flow j, $j \in \{1...A\}$. By the conditional variance formula [38] we have:

$$\text{Var}(P) = E[\text{Var}(P|A)] + \text{Var}(E[P|A]). \tag{6}$$

[9]For the proof of this theorem we will be considering flows that have the same round-trip times. However, as we will see in Section 7 Equation (5) is, in practice, remarkably accurate even if this is not the case.

Since there are no drops, the W_j's ($j \in 1...A$) are independent of the random variable A. It is then easy to see that:

$$E[\text{Var}(P|A)] = E[A]\text{Var}(W), \tag{7}$$

and:

$$\text{Var}(E[P|A]) = (E[W])^2\text{Var}(A). \tag{8}$$

Now, recall from Section 4 that $\sigma^2 t^{2H}$ is the variance of the amount of traffic that arrives at the queue in the interval $(0, t]$. As in Section 4 denote this amount of traffic by $A(t)$, and let $N(t)$ be the number of time-slots elapsed by time t. We can write $A(t) = \sum_{i=1}^{N(t)} P(i)$, where $P(i)$ is the random variable representing the number of packets arriving at the queue within slot i.

In steady-state the $P(i)$'s are identically distributed. Accounting for long-range dependence in the sequence $\{P(i), i = 1, 2, ..., N(t)\}$, we can write $\text{Var}(A(t)) = (N(t))^{2H}\text{Var}(P) = \sigma^2 t^{2H}$. Now, for t large enough $N(t) = \frac{t}{E[RTT]}$, and hence:

$$\sigma^2 = \frac{\text{Var}(P)}{(E[RTT])^{2H}}. \tag{9}$$

From Equations (6)...(9) we get Equation (5).

□

Notice that Equation (5) can be used in predicting the variance $\sigma^2 t^{2H}$ at time-scales t larger than the average round-trip time $E[RTT]$. Recall that Equation (4) is accurate for such time-scales. We will discuss about the impact of TCP traffic dynamics at smaller time-scales in Section 7.

Now, recall that $E[A]$ and $\text{Var}(A)$ in Equation (5) are known quantities. Hence, what remains to complete the calculation of σ^2 is to compute $E[W]$, $\text{Var}(W) = E[W^2] - (E[W])^2$, and $E[RTT]$.

We begin by $E[RTT]$. Let $E[D]$ be the average number of round-trips that a flow needs in order to complete. Using Little's Law we can write:

$$E[RTT] = \frac{E[A]}{rE[D]}. \tag{10}$$

Since $E[A]$ and r are known quantities, we only need to find $E[D]$.

Recall that S is the random variable that represents the size of a flow. Now, suppose that the maximum window size of a flow is W_{max}. We divide flows into two categories: (i) short flows, whose size is less than or equal to $2W_{max}$, and (ii) long flows whose size is larger than $2W_{max}$. Given TCP's AIMD (Additive Increase Multiplicative Decrease) mechanism, this separation implies that a short flow spends its lifetime in slow start, and may send W_{max} packets at most once during its lifetime. We can write:

$$E[D \mid \text{short flow}] = E[\lfloor \log_2 S \rfloor + 1_{[S - \sum_{i=0}^{\lfloor \log_2 S \rfloor - 1} 2^i > 0]} \mid S \le 2W_{max}], \tag{11}$$

where $1_{[.]} = 1$ if the condition in the brackets is satisfied, and 0 otherwise. Now, long flows spend approximately $\log_2 2W_{max}$ round-trip times in slow-start and then send W_{max} packets per round-trip for the rest of their lifetime. Hence:

$$E[D \mid \text{long flow}] = E[\lfloor \log_2 2W_{max} \rfloor + \lfloor \frac{S - \sum_{i=0}^{\lfloor \log_2 2W_{max} \rfloor - 1} 2^i}{W_{max}} \rfloor + 1_{[R(S)>0]} \mid S > 2W_{max}], \tag{12}$$

where:
$R(S) = S - \left[\sum_{i=0}^{\lfloor \log_2 2W_{max} \rfloor - 1} 2^i + \lfloor \frac{S - \sum_{i=0}^{\lfloor \log_2 2W_{max} \rfloor - 1} 2^i}{W_{max}} \rfloor W_{max}\right]$.
Notice that $0 \leq R(S) < W_{max}$. We refer to $R(S)$ as the residual data of the flow. Since we know $F(s)$, we can compute and uncondition the expectations above and find $E[D]$. Thus, we can now compute $E[RTT]$ using Equation (10). (Note that the analysis can be extended for the case where flows do not have a common W_{max}, however we need to knowledge of the probability distribution of W_{max}. Here, we have used a common W_{max} for all flows, based on the simplifying assumption that most of the end hosts will be using their default TCP maximum window size, which is the same for the majority of major operating systems, *e.g.* 32KB.)

Since we know the expected flow size and the expected number of rounds a flow needs to complete, it is easy to see that the average window size of a flow is: [10]

$$E[W] = \frac{E[S]}{E[D]}. \quad (13)$$

What remains, is to compute the mean square window size of a flow $E[W^2]$. For this, we first need to find an expression for the expectation of the sum of the squares of the window sizes that a flow reaches during its lifetime. We denote this expectation by $E[S^*]$. Considering TCP's AIMD mechanism as we did before, and distinguishing again short and long flows we can write:

$$E[S^* \mid \text{short flow}] = E[\sum_{i=0}^{\lfloor \log_2 S \rfloor - 1} (2^i)^2 + (S - \sum_{i=0}^{\lfloor \log_2 S \rfloor - 1} 2^i)^2 \mid S \leq 2W_{max}], \quad (14)$$

$$E[S^* \mid \text{long flow}] = E[\sum_{i=0}^{\lfloor \log_2 2W_{max} \rfloor - 1} (2^i)^2 + \lfloor \frac{S - \sum_{i=0}^{\lfloor \log_2 2W_{max} \rfloor - 1} 2^i}{W_{max}} \rfloor (W_{max})^2 + (R(S))^2 \mid S > 2W_{max}], \quad (15)$$

where $R(S)$ is the residual data of a flow as defined earlier. As before, knowing $F(s)$, we can uncondition these expectations and find $E[S^*]$. The relation for $E[W^2]$ is given in the following lemma:

LEMMA 1.

$$E[W^2] = \frac{E[S^*]}{E[D]}, \quad (16)$$

where $E[S^*]$ and $E[D]$ as defined earlier.

PROOF. Assume again that the time is slotted with the duration of the current slot equal to the current round-trip time. Now, let Y be the sum of the squares of the window sizes of all active flows, i.e. $Y = \sum_{j=1}^{A} W_j^2$. As before, since there are no drops the W_j's ($j \in 1...A$) are independent of the random variable A. We can write:

$$E[Y] = E[W^2]E[A]. \quad (17)$$

Let $N(t)$ be the number of time-slots elapsed by time t as before, and denote by $F(t)$ the total number of flows that have completed service within $N(t)$ slots. The average number of rounds for a flow to complete can be expressed as $E[D] = \lim_{t\to\infty} \frac{\sum_{i=1}^{N(t)} A(i)}{F(t)}$, where $A(i)$ is the number of active flows in slot i. Also, the average number of active flows in a slot can be written as $E[A] = \lim_{t\to\infty} \frac{\sum_{i=1}^{N(t)} A(i)}{N(t)}$. From the last two equations we get that $\lim_{t\to\infty} \frac{N(t)}{F(t)} = \frac{E[D]}{E[A]}$. Further, it is easy to see that $E[S^*] = \lim_{t\to\infty} \frac{\sum_{i=1}^{N(t)} \sum_{j=1}^{A(i)} (W_j^i)^2}{F(t)}$, where W_j^i is the congestion window size of flow j ($j \in 1...A(i)$). And, we can write $E[Y] = \lim_{t\to\infty} \frac{\sum_{i=1}^{N(t)} \sum_{j=1}^{A(i)} (W_j^i)^2}{N(t)}$. Therefore, (from the last three relationships) we can conclude that:

$$E[Y] = \frac{E[S^*]}{E[D]} E[A]. \quad (18)$$

Combining Equations (18) and (17) we get the result. □

We have now computed all the parameters required to estimate σ^2.

7. EXPERIMENTS

In this section we use the ns-2 simulator [27] to validate our theoretical arguments and to demonstrate the procedure for efficiently identifying uncongested links when performing topology downscaling. In particular, we present two sets of experiments. In the first set we consider a single link shared by TCP flows, in order to verify the accuracy of the model of Section 4 and of our parameter estimation (Section 6), as well as to give insights on the queueing behavior of Internet links that are shared by a large number of flows. In the second set, we use the topology of the CENIC backbone [5], to demonstrate the procedure for identifying uncogested links when performing topology downscaling on real networks.

7.1 Single Link Experiments

Let $N \geq 1$ be a scaling factor. We consider a single link/queue like the one shown in Figure 4, having capacity NC, propagation delay $Tprop$, and buffer size $B = 2NCTprop$ (i.e. equal to the bandwidth-delay product). TCP flows arrive at the link at random times, according to a Poisson process, with rate $Nr = N95$ flows/sec. (Of course, while flow arrivals are Poisson, packet arrivals are dictated by the TCP dynamics. Further, similar results hold for any other flow arrival process.) The number of data packets S in each flow follows a bounded Pareto distribution with average $E[S] = 11.5$packets, maximum 10^6packets, and shape parameter $\alpha = 1.34$. The size of an IP data packet is 1040 bytes, $T_{prop} = 50$ms, and $C = 10$Mbps. Finally, $W_{max} = 20$packets and the simulation time is 10000sec. We study the queueing dynamics of the link as N increases, i.e. as if this was a backbone link. (Notice that the offered load is $\rho = \frac{NrE[S]}{NC} = \frac{rE[S]}{C} = 0.91 < 1$, and does *not* change as we vary N. This why we have chosen to scale the flow arrival rate r with exactly the same factor N that the link capacity C increases.)

We start by verifying that the aggregate packet arrival process at the link can be approximated by a Gaussian distribution. Figures 5(i) and 5(ii) show that this is indeed the case, even for N's as small as 1 and 6 respectively. Note that for $N = 1$ the average number of active flows is approximately $E[A] = 40$, and the packet drop ratio is around 1.2%. This implies that the Gaussian approximation is accurate even when the number of multiplexed flows is relatively

[10] A formal proof for this relation goes along the same lines with the proof of Lemma 1, which we will state shortly.

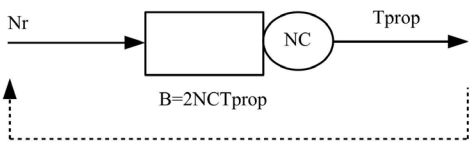


Figure 4: Single link topology.

small and there are packet drops. This is in agreement with the observations in [2]. [11] For $N = 6$, the average number

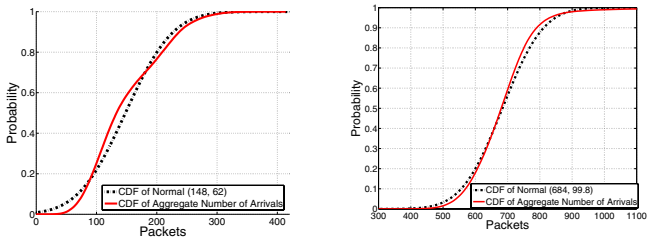


Figure 5: The commulative distribution function (CDF) of the sum of the aggregate number of arrivals passing through the router during a round-trip time, and its approximation with a Gaussian CDF with the same parameters: (i) N=1, and (ii) N=6.

of active flows is $E[A] = 162$, and the percentage of dropped packets 0.02%. In this case, because there are more flows active in the system, the Gaussian approximation is more accurate. This is evident from Figure 5(ii). Also, notice that the drop ratio is smaller than the case where $N = 1$. This is in agreement with the model of Section 4, which implies that for any level $\delta > 0$, as N, and hence B, increases, the probability that the buffer content exceeds δB decreases.

We now test the accuracy of the model of Section 4 and of the expressions derived in Section 6. Recall, that for the purposes of downscaling we are interested in identifying which of the links that do not impose packet drops are uncongested, i.e. impose negligible queueing delays. As we have observed from the simulator, drops stop occurring for $N > 10$. Thus, we show results for $N = 11, 16$, and 32.

We estimate λ, σ^2 and H, using the formulas of Section 6. Recall, that in order to compute σ^2 we also need estimates for $E[A]$ and $\mathrm{Var}(A)$. These are extracted from the simulator. We compute the rest of the required parameters, and their values are: $E[D] = 2.65$rounds (which gives $E[W] = 4.3$packets), and $E[S^*] = 127.5$packets (which gives $E[W^2] = 48$packets). ($E[RTT]$ is computed by Equation (10) given the corresponding value for $E[A]$ and the flow arrival rate.)

Table 1 gives the values for $\lambda, E[A], \mathrm{Var}(A)$ and the resulting σ^2, as we vary N. In all cases $H = 0.83$ (as the shape parameter of the flow-size distribution remains the same).

N	λ(pkts/sec)	$E[A]$	$\mathrm{Var}(A)$	σ^2 $(\mathrm{pkts/sec})^2$
11	12018	281	578	858148
16	17480	404	644	1093018
32	34960	807	929	1864839

Table 1: Flow- and packet-level statistics at the link.

Figure 6 shows that the model is quite accurate for approximating the queue length distribution, especially for large N, as expected, and also verifies that our parameter estimation is correct. (The latter has been also verified by comparing the derived theoretical values with the corresponding simulation values.) The plots also validate the argument that in backbone links, where N is sufficiently large, queueing delays can be ignored. Indeed, for $N = 32$ the average queueing delay is approximately $\overline{T} = 1$ms, which is two orders of magnitude smaller than the two-way end-to-end propagation delay of a packet (which is 100ms). And, this is the case even for links working at above 90% utilization, like the one in this example.

This last observation motivates us to study the amount of multiplexing (value for N) required at different offered loads ρ, such that for the majority of time, the queuing delay T remains below a sufficiently small fraction of the end-to-end propagation delay. This is important for topology downscaling, where we can ignore links with negligible queueing delays (compared to the end-to-end delays).

Figure 7 shows the value for N such that the queueing delay is one order of magnitude smaller than the end-to-end propagation delay for at least 90% of the time, for different values of ρ. From the figure we observe that at small offered

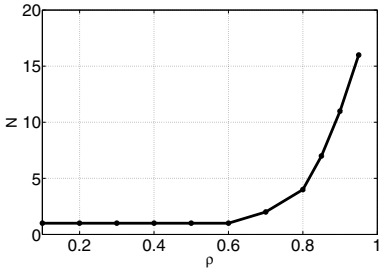


Figure 7: The value for N such that $P(T < 0.1 \times 2Tprop) > 0.9$, as a function of the offered load ρ.

loads, a small value for N is sufficient. In particular, for $\rho \leq 60\%$ $N = 1$ is sufficient, whereas as ρ increases, N also increases as expected, with the increase being faster than exponential as $\rho \to 1$. For example, for $\rho = 90\%$, we need $N = 15$. Even in this case however, this corresponds to a flow arrival rate of $95N = 1425$flows/sec, a capacity of $10N = 150$Mbps, a buffer size of $120N = 1800$packets (or 15Mbits), and a corresponding average number of active flows $E[A] = 387$. These values are still quite smaller than the usual values for backbone networks, whose links typically have capacities 2.5Gbps – 10Gbps, buffer sizes 625Mbits – 2.5Gbits, and carry more than 10000 active flows, *e.g.* see [2].

[11] The study in [2] verified the Gaussian approximation assumption for the case of a single link that is shared by long-lived persistent TCP flows with unbounded window sizes, operating at 100% utilization. Here, we verify this for the more realistic case where TCP flows arrive at random times, have random sizes, bounded windows, and $\rho < 1$.

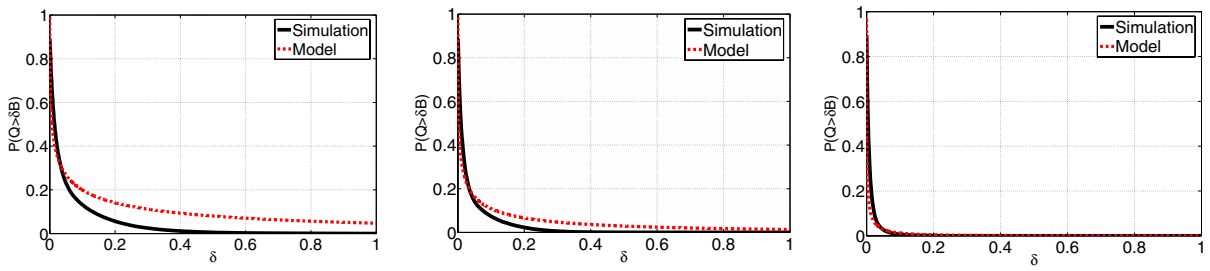


Figure 6: Queue exceedance probability $P(Q > \delta B)$ against the buffer level δ: (i) $N = 11$, (ii) $N = 16$, and (iii) $N = 32$.

Finally, note that we have verified that the model of Section 4 is quite applicable for all link utilizations above 65%. At smaller utilizations we have observed that it may underestimate the true queue occupancy for small δ's (greater than zero). This agrees with experimental observations in [34], and it is due to the fact that the model cannot capture TCP traffic dynamics at small time-scales (smaller than the round-trip time), whose effect is more evident at low link utilizations [10]. (To capture traffic dynamics at small time-scales one would need to use an extension of the model of Section 4, *e.g.* like the one derived in [10], which also requires parameter estimation for small time-scales.)

However, for the purposes of downscaling such discrepancies do not affect our decisions of whether to keep or ignore a link, since as in [31, 32], we are usually interested in *order-of-magnidute* comparisons between queueing delays and end-to-end delays, and therefore, the exact absolute queueing delay value of a link is not important. In addition, one can argue that backbone links at utilizations below 50% impose insignificant queueing, and we can always consider them as uncongested [10, 34], without the need of using the model to approximate their queue distribution.

Due to space limitations, we do not present more results for this simple topology. For results at other link utilizations we refer the interested reader to [30].

7.2 Cenic Backbone Experiments

We now consider the topology of the CENIC backbone [5], which is shown along with link information in Figure 8. Note that the CENIC maps do not include information about the propagation delays of the links and the paths of the packets that traverse them. We estimate the propagation delay of a link by dividing the length of the link over the propagation velocity of the signal (taken as 133000 miles/sec). The propagation delay for all the links that belong to the same geographic area is taken as 0.1ms and for the rest of the links is shown in Figure 8 (appended next to each link). Further, the buffer size of each link equals the bandwidth-delay product, where the delay factor is taken equal to the maximum end-to-end propagation delay of a flow, which is approximately 10ms.

We let each possible source-destination pair in the topology to correspond to a group of flows. (Notice that links are bidirectional.) Hence, in total there are 600 groups of flows. The flow arrival rate for all groups of flows is 100flows/sec, except from the group that enters *SVL(dc1)* and exits *SVL(hpr)*, whose rate is 5200flows/sec, and from

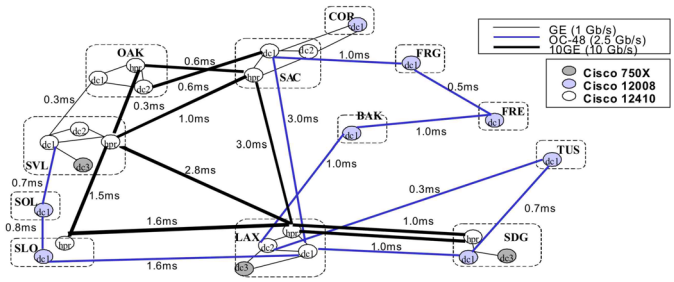


Figure 8: The CENIC Backbone.

the group that enters *SVL(hpr)* and exits *LAX(hpr)*, whose rate is 90000flows/sec. The larger flow arrival rate for these two groups forces the offered load on link *SVL(dc1)-SVL(hpr)* to be approximately 88% and on link *SVL(hpr)-LAX(hpr)* to be 95%.

Link *SVL(dc1)-SVL(hpr)* imposes packet drops, and hence it is congested. No other link in the topology imposes packet drops. We are interested in studying the performance of the congested link and of the groups of flows that traverse it (which, as before, are called groups of interest). According to [31, 32] one can build a scaled replica consisting of this link along with the groups of interest. And, for performance prediction to be accurate, the scaled replica should also include all other congested links in the topology that these groups traverse. Hence, since we know that no other link imposes drops, our task is to identify if there are links traversed by groups of interest that have significant delays, and if so, include them in the scaled replica. Before proceeding, we review and summarize the general procedure that we follow.

Procedure for identification of uncongested links: (i) From the network topology and routing information, we identify and ignore every link for which the traffic it carries is being forwarded from/to links for which the sum of their capacities is smaller that the capacity of the link. (See Section 2). (ii) For all other links we use a flow-level measurement tool, *e.g.* such as NetFlow [24], to estimate: (a) the flow-size distribution, (b) the flow arrival rate, and (c) the average, and the variance of the number of active flows. (iii) For each of these links, we use Equations (3)...(5) to compute λ, H, and σ^2. (iv) We use the model of Section 4 (Equations (1) and (2)) to approximate the queue distribu-

tion on each of these links. (v) From the network topology and traffic matrix we calculate for each of these links the average *two-way* end-to-end propagation delay among the groups of flows that traverse them, and (vi) as in [31, 32] we ignore all those links for which their average queueing delay is one order of magnitude smaller than the corresponding two-way end-to-end propagation delay.

Note, that a rule-of-thumb to expedite the above procedure is to measure, after step (i), the offered load ($\rho = \frac{rE[S]}{C}$) on all remaining backbone links and directly ignore all links for which this load is quite low, *e.g.* $\rho \leq 50\%$. As mentioned before, this is based on the observation that such links are always uncongested, and this is the case for the majority of the links [10, 34].

As mentioned before, the offered load on link *SVL(hpr)-LAX(hpr)* is approximately 95%. This link is being traversed by a total of 102 groups of flows, out of which 37 are groups of interest. The offered load on all other links that are traversed by groups of interest is below 40%. Following our procedure, and using the aforementioned rule-of-thumb, the only link that we need to approximate the queue length distribution to decide whether it is congested, is link *SVL(hpr)-LAX(hpr)*. (All other links are ignored since they are uncongested.) The total average flow arrival rate at this link is $r = 95150$flows/sec, the flow characteristics are the same as before (except that $E[S] = 12$packets), the average number of active flows on the link is $E[A] = 1482$, and its variance is $\text{Var}(A) = 2464$. These parameters are extracted from data obtained from the simulator at the edges of the network. As before, we can compute $\lambda = 1141800$packets/sec, $\sigma^2 = 629752212(\text{packets/sec})^2$, and $H = 0.83$. Note that while a small proportion (11%) of the active flows on link SVL(hpr)-LAX(hpr) may experience some drops on the congested link *SVL(dc1)-SVL(hpr)*, the impact to the accuracy of our parameter estimation, which assumes that flows do not experience drops, is insignificant. (Notice that it is not unrealistic to expect that only a small proportion of active flows on a backbone link will experience drops elsewhere along their path, given that backbone links carry thousands of flows and that the number of concurrent congested links is usually small.) Figure 9 shows how accurately we can approximate the queue length distribution.

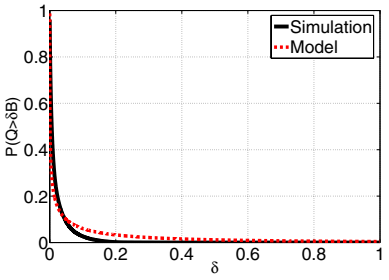


Figure 9: Queue exceedance probability $P(Q > \delta B)$ against the buffer level δ for link *SVL(hpr)-LAX(hpr)*.

Our approximation yields an average queueing delay of $\overline{T} = 0.26$ms and the actual is $\overline{T} = 0.17$ms. In both cases this is one order of magnitude smaller than the average end-to-end propagation delay of flows that traverse the link under study, which is 6.33ms. Therefore we ignore this link.

Notice that despite the fact that the utilization of the above link was so high (approximately 95%), both the high degree of statistical multiplexing and the large capacity have rendered the link uncongested. Without sophisticated tools, *e.g.* like the large deviations theory we are utilizing here, it would be very hard to realize this based only on simple metrics like the link's utilization.

To validate the effectiveness of the whole procedure we use DSCALEd [31, 32] (which accounts for the missing uncongested links by imposing appropriate delays at the sources of the packets), to build a scaled replica consisting of the congested link *SVL(dc1)-SVL(hpr)* only, and the groups of interest (71 in total). In Figure 10 we present some of the most important performance metrics that we can predict using the scaled replica, and we compare them to that of the original system (Figure 8). In particular, we show the distribution of the number of active flows on link *SVL(dc1)-SVL(hpr)* in the original and scaled system, and the end-to-end flow delay histograms of two groups of interest, which we refer to as *grp1* and *grp2*. (The link *SVL(hpr)-LAX(hpr)*, which we previously decided to ignore, belonged to the path of *grp2* in the original system.)

It is visually evident from the plots that performance prediction is quite accurate, despite the fact that we went from an original topology of 41 links to a downscaled topology consisting of a single link only!

Further, in addition, if we use the same well-known statistical measure to quantify differences between two distributions as in [32], i.e. the Histogram Similarity Measure (HSM), we find that the average HSM for these plots is 0.85, which is quite high, as in [32]. (The HSM is defined as $HSM = 1 - C_v$, where $C_v = \sqrt{\frac{\chi^2}{2}}$ is the Cramer's V coefficient, and χ^2 is the well-known chi-square statistic [36]. HSM=1 means that two distributions are identical). Therefore, the proposed procedure can be efficiently applied to identify and ignore uncongested links.

8. RELATED WORK

We now review related work on the applicability of the model of Section 4, and on estimating σ^2. For related work on network downscaling see [32].

As mentioned earlier, the model presented in Section 4 has been derived in several studies and its effectiveness has been verified in the context of open-loop networks, *e.g.* see [8, 26, 14]. Its applicability has been also demonstrated for Internet backbone traffic, *e.g.* see [42, 34]. And, it has been used in this later context by authors for their theoretical arguments, *e.g.* in [40, 7].

In this study we have shown that this model can be also effectively applied in the context of topology downscaling. Further, we have clearly identified the necessary conditions for the model to be applicable, and we have used ns-2 simulations with TCP traffic to further validate it.

In contrast to earlier studies that have utilized the model by extracting its parameters from packet-level traces, *e.g.* [42, 34], in this study we have chosen to infer this information from flow-level statistics. In the process, we derived a formula that relates the variance σ^2 of the packet arrival process to some flow-level information. The most relevant to this are the studies in [3, 4, 16]. We now explain the main differences of our approach.

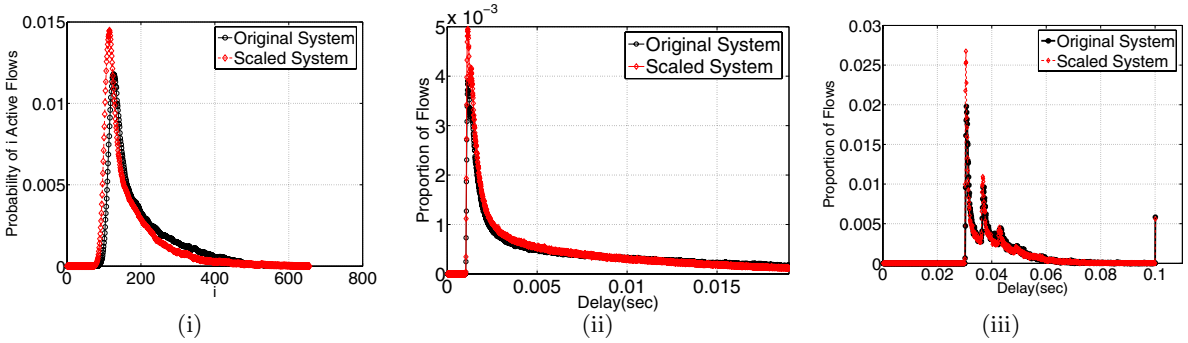


Figure 10: (i) Distribution of the number of active flows on the congested link *SVL(dc1)-SVL(hpr)*, (ii) *grp1* end-to-end flow delay histogram, and (iii) *grp2* end-to-end flow delay histogram.

First, for their formula derivation, all of these studies have assumed flows that arrive to the system according to a Poisson process. In addition, in [16] the author has also assumed a bufferless link model and modeled the number of active flows as an $M/G/\infty$ queue (which is not accurate when queueing delays are not exactly equal to zero). During our formula derivation, none of these simplifying assumptions have been made. Further, in [3, 4] the notion of "shots" was introduced to describe how flows transmit their packets. To accurately estimate the variance requires correct estimates for the shapes of the shots. And, for correctly estimating the shapes, in general, requires further measurements. Also, in [16] it is assumed that the packets of a flow are spread uniformly in time. In contrast, in our study we have not made any assumptions on how flows transmit their packets. We have explicitly taken into consideration TCP's AIMD mechanism and long-range dependence.

Finally, the study in [3, 4], which is the most relevant, derives a variance formula that requires (in addition to the flow arrival rate) knowledge of the expectation $E[\frac{S^2}{T}]$, where S is the flow size and T the flow duration. The authors do not provide an analytical expression for this expectation, but instead, compute it offline using measured data. However, as the authors agree, an efficient implementation of their model would require an online estimation of this expectation, which, in practice, requires to keep track of all flow sizes and their corresponding durations. The complexity of such a task is not trivial, especially in high-speed backbone networks.

In our study, we still require knowledge of the flow sizes, but we do not need to keep track of the corresponding durations. Instead, we only need estimates on the first two moments of the number of active flows on a router, which can be easily estimated online (for example, as suggested in [3, 4] for some other metrics, by using appropriate weighted moving algorithms, like the ones used by TCP to estimate the average round-trip time and its variation), independently from the flow sizes or any other quantity.

9. CONCLUSION AND FUTURE WORK

This paper complements recent work on topology downscaling of Internet-like networks [31, 32]. In particular, this paper proposes a procedure to identify links with negligible queueing delays that can be ignored when building scaled-down replicas.

This study also goes beyond the context of network downscaling. It demonstrates how a well-known model from the large-deviations theory can be efficiently utilized in practice, and it presents a new simple formula that relates the variance of the packet arrival process to flow-level statistics.

Future work consists of further validating the proposed procedure using more real network topologies, in both simulation and emulation environments. One of our main goals is to further understand through these experiments, the range of values that a network operator could use as a queue-size threshold, to decide, in practice, which links to ignore. Another interesting future work direction is to consider UDP traffic, in addition to TCP, as well as TCP traffic dynamics at small time-scales.

Further, note that this work (and the work in [31, 32]) is about scaling down Internet's topology to preserve *application level* metrics. Recently, a new line of work has suggested that one could build topology replicas that preserve most of the *graph level* properties [21]. It would be very interesting to investigate if these two lines of research can be combined, to build scaled down replicas, where one could study both application level metrics as well as how these might be affected by the underlying graph structure and the associated routing protocols.

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